SERVICE MANUAL

USA Instruments, Inc.

Document 2383477-8

Revision 2

GE Signa® 3.0T
MILLENNIUM III
4-CHANNEL NEUROVASCULAR (NV) ARRAY

GE Catalog Part Number: M1385AZ (Eclipse™)
M1385AN (EXCITE™)

Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
  http://data.supportcentral.ge.com/upload/15325/doc_324531.doc
- Fill out a report on http://3.28.216.127/sctq/InstallFulfill/IFHome.htm
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

Language Policy For Service Documentation (Dir. 2128126)

WARNING

- This service manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

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- Le non–respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

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- Falls ein fremder kundendienst eine andere sprache benötigt, ist es aufgabe des kunden für eine entsprechende übersetzung zu sorgen.
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- Este manual de servicio sólo existe en inglés.
- Si algún proveedor de servicios ajeno a gems solicita un idioma que no sea el inglés, es responsabilidad del cliente ofrecer un servicio de traducción.
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- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
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- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione esterno alla gems richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell’apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Non tenere conto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all’addetto alla manutenzione, all’utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

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・このサービスマニュアルは英語版しかありません。

・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

・このサービスマニュアルを熟読し、理解せずに装置のサービスを行わないでください。

・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

TABLE OF CONTENTS

SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the 4-Channel Neurovascular Array, refer to the coil label (see *Figure 1*).

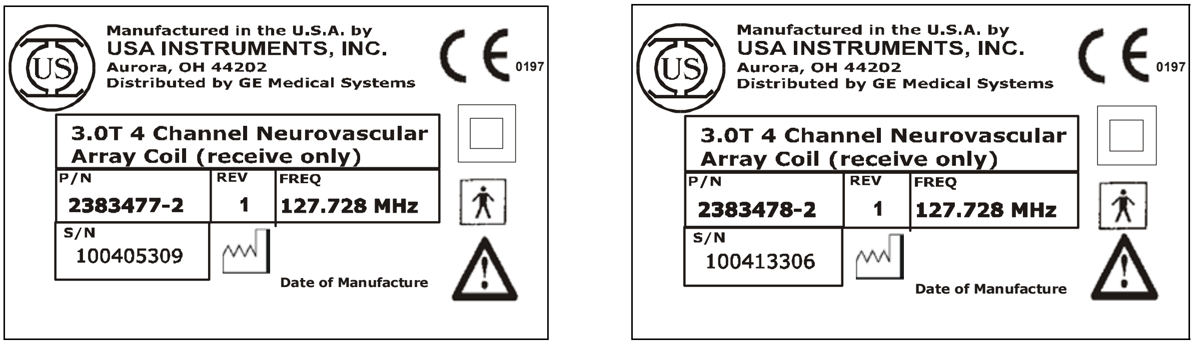


Figure 1: Coil labels.

Figure 2 shows a picture of the 4-Channel Neurovascular Array.

Figure 2: The 4-Channel Neurovascular Array.

ECLIPSE SHIPPING LIST – TABLE 1-1A

Box #	Part Name	GE Part #	Qty
1	Coil	2383477-2	1
2	Phantom Kit	2383477-3	1
2	Phantom Positioner	2383477-4	1
1	Patient Comfort Pad	2383477-5	1
1	Wedge Pads	2383477-6	2
1	Operator’s Guide	2383477-7	1
1	Service Manual	2383477-8	1

EXCITE SHIPPING LIST – TABLE 1-1B

Box #	Part Name	GE Part #	Qty
1	Coil	2383478-2	1
2	Phantom Kit	2383477-3	1
2	Phantom Positioner	2383477-4	1
1	Patient Comfort Pad	2383477-5	1
1	Wedge Pads	2383477-6	2
1	Operator's Guide	2383477-7	1
1	Service Manual	2383477-8	1

1-2 Compatibility

This coil is compatible with the GE Signa® 3.0T Eclipse or the GE Signa® EXCITE system.

Signa® 3.0T Eclipse, Coil Part Number 2383477-2.

Signa® 3.0T EXCITE, Coil Part Number 2383478-2.

1-3 Related Documentation

Operator's Guide, GE Part Number 2383477-7.

Signa® 3.0T Eclipse Service Methods CD, GE Part Number 2360530.

Signa® 3.0T EXCITE Service Methods CD, GE Part Number 2389160.

1-4 Environmental Requirements

Storage Requirements

The coil should be stored in the scanner room.

Dimensions

25.6" x 18.7" x 13.8" (65.0 cm x 47.5 cm x 35.0 cm)

Weight

20 lbs. (9.1 kg)

1-5 Theory of Operation

The 4-Channel Neurovascular Array is designed for imaging the brain, cervical spine, soft tissues, and vasculature of the head, neck and upper chest extending to the cardiac region. The coil consists of three sections: a main coil base, a split-top designed head former, and a detachable chest plate.

Coupling of the coil to the transmit B1 field is prevented by using RF choking circuits or RF chokes. Coil contains seven elements; saddle coils #1 - #5, and loop coils #1 - #2. These seven coil elements form three quadrature pairs labeled as Channel 1 (Saddles #1 and #2), Channel 2 (Saddles #3 and #4), Channel 3 (Loop #1 and Saddle #5) and one linear channel labeled as Channel 4 (Loop #2). The RF chokes are switched both actively and passively using PIN diodes and small-signal diodes. The PIN diodes are turned on by passing a DC current (200 mA) to each of the PIN diodes while the small-signal diodes are turned on by the induced RF voltage in the chokes, coupled from the transmit field (passive decoupling). Each small-signal diode is turned on when the induced RF voltage reaches about 0.5 Volts. When the PIN diodes and small-signal diodes are turned on, the RF choking circuits become very high impedance (typically above 2 KΩ) blocks, compared to the other circuit elements (less than 25 Ω). These high impedance elements segregate the coil circuitry into several isolated electrical segments. Therefore, the deresonated coil does not support any RF current flow in its circuit that might be induced by the body coil transmission. In the event that one of the diodes fails, it will fail in a shorted mode and permanently deresonate the coil (fail safe).

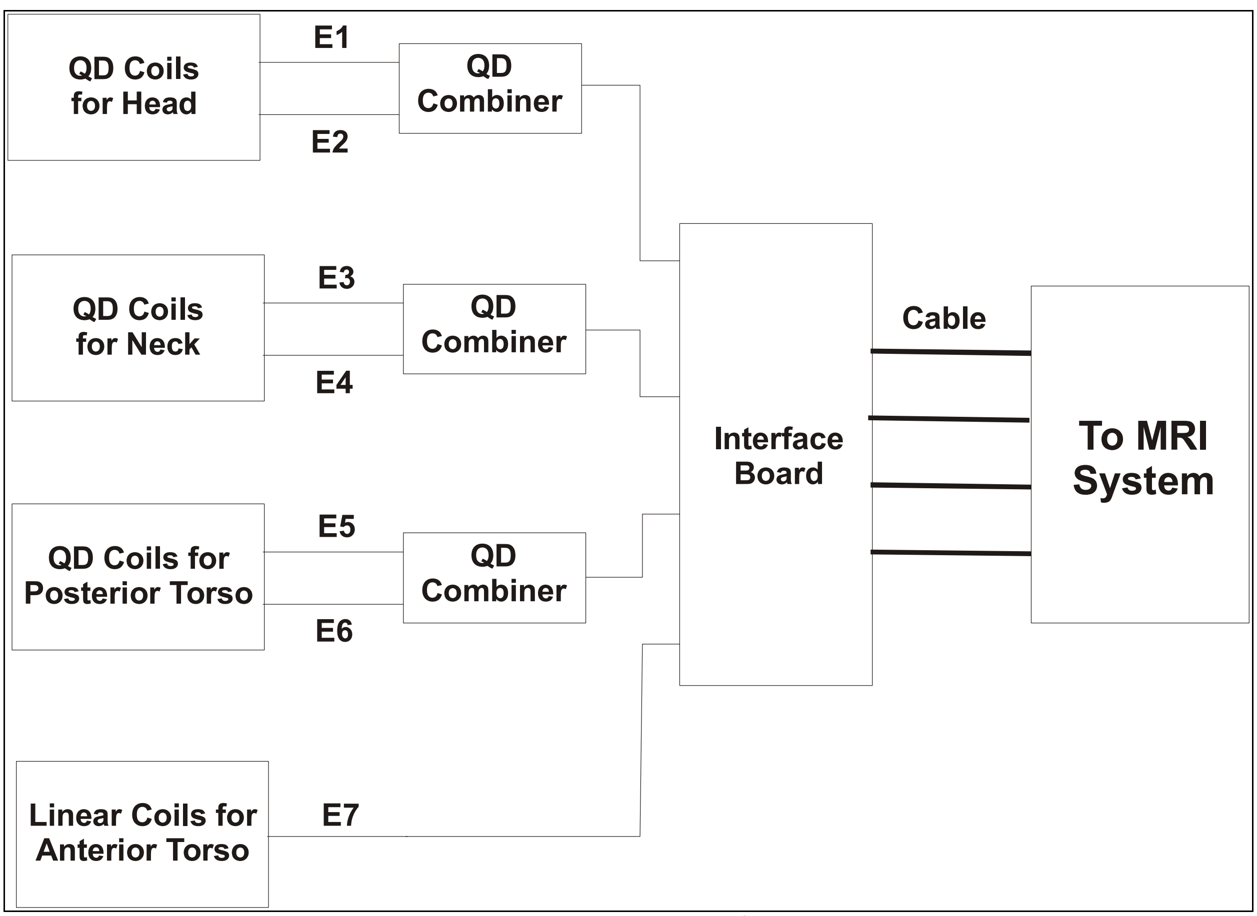


Figure 3: Block diagram of the coil.

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration

No configuration should be required. For reference, the configuration information is included in the Appendix; however, that configuration is only current as of the printing of this manual.

2-2 Installation Functional Checks

1. From the Scan Desktop, start new scan by selecting [**New Pt**]; set **Patient ID** to "*geservice*" and **Patient Weight** to "*111*" pounds. Click [**Patient Position**] to open protocols window.
2. Plug the coil into the patient table.
3. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1A

Description	GE Part #	Qty
Phantom Kit	2383477-3	1
Phantom Positioner	2383477-4	1

3-2-2 Explanation of Procedure

The 4-Channel Neurovascular Array has two sections controlled by the coil configuration files: For Eclipse 3.0T systems the coil configuration names are 4CH_NVARRAY and 4CH_NVHEAD. For EXCITE 3.0T systems, the coil configuration names are US4NVFULL and US4NVHead. For evaluation of coil performance, coronal images will be acquired on the Head and chest sections of the coil using the protocol given below.

SNR measurements must be made for the Eclipse, 4CH_NVARRAY or Excite, US4NVFULL mode, requiring one set of signal and noise scans. Refer to the Data Sheet in Appendix 7-1 to understand the data required to calculate the composite SNR for the Head and Chest section in the Eclipse 4CH_NVARRAY or Excite US4NVFULL mode of operation. All ROI measurements are made on the composite image.

The image quality check uses two different protocols for signal and noise image acquisition. The signal scan is an **<u>FSE</u>** sequence used to minimize susceptibility and B_0 inhomogeneity effects. The noise scan is a **<u>GRE</u>** sequence that has a Control Variable (do_noise) to eliminate the transmit RF completely during the scan. The signal scan **<u>must</u>** be run prior to the noise scan as the R1, R2, and TG values from the signal scan are used for the noise scan.

3-2-3 Signal Scan

1. From the **Scan Desktop**, start new scan by selecting [**New Pt**]; set **Patient ID** to "*geservice*" and Patient Weight to "*111*" pounds. Click [**Patient Position**] to open protocols window.

2. Position the phantom set on the coil without the patient pad, using the positioner (see *Figure 4*).

3. Landmark on the horizontal line corresponding to the region of interest (see *Figure 4*).

Figure 4: Positioning the phantoms on the coil.

4. At the magnet, press “**Alignment Light**” to turn on the light. Move the cradle to align the coil to the alignment lights. Press “**Landmark**” to landmark the alignment.

5. Move the coil to scan position by clicking on the “Move to Scan” button.

6. At the console, verify the coil has been properly identified by the system: correct picture on the screen and name in the coil field. If the system does not recognize the coil, refer to Section 2 - Setup and Calibration in the Service Manual.

7. Set the protocols per the Signal section from Table 3-2-4 Signal and Noise Protocols.

8. Select [**Save Series**] the [**Prepare to Scan**].

9. Prior to the scan, open the [**Display CVs**] menu under [**Research Operations**].

10. Set the “**saveinter**” CV to 1, saving the intermediate images. This is required to ensure all channels are operating well.
11. Select [**Auto Prescan**].
12. Record the R1, R2 and TG values obtained, then select [**Scan**].
13. If not continuing to noise scan, be sure to end the exam when finished. This will ensure that all CVs return to their default values.

3-2-4 Noise Scan

A signal scan must be run **prior** to the noise scan as the same R1, R2 and TG values must be used for both the signal and noise scans. Do **not** run an Auto Prescan prior to the noise scan as the values will be changed.

1. Copy the signal scan series. Use [**Copy Series**] (highlight signal series and click right mouse button) and [**Paste Series**] in **RX Manager**.
2. Click [**View Edit**] and set the protocols per the Noise section from Table 3-2-4 Signal and Noise Protocols.
3. Click [**Save Series**] and click [**Prepare to Scan**].
4. Open the [**Display CVs**] menu under [**Research Operations**].
5. To obtain the noise images, set the "**rhformat**" and "**do_noise**" CVs to 1.
6. Start [**Manual Prescan**], look at the power spectrum and verify R1, R2, TG and Bandwidth have not changed from the previously completed signal scan.
7. Select [**Done**] and exit without changing any parameters.
8. Select [**Scan**].

SIGNAL AND NOISE PROTOCOLS – TABLE 3-2-4

	Eclipse		EXCITE	
	Signal	**Noise**	**Signal**	**Noise**
Patient/Exam Information				
Patient ID	geservice	geservice	geservice	geservice
Patient Name	4CHNV Array	4CHNV Array	4CHNV Array	4CHNV Array
Patient Weight	111 lbs. (50 kg)	111 lbs. (50 kg)	111 lbs. (50 kg)	111 lbs. (50 kg)
Patient Position	Supine	Supine	Supine	Supine
Patient Entry	Head First	Head First	Head First	Head First
Coil	4CH_NV ARRAY	4CH_NV ARRAY	US4NVFULL	US4NVFULL
Series Description	Signal	Noise	Signal	Noise
Imaging Parameters				
Plane	Coronal	Coronal	Coronal	Coronal
Mode	2D	2D	2D	2D
Pulse Seq	FSE	GRE	FSE-XL	GRE
Imaging Options	none	none	none	none
PSD Name	*leave blank*	*leave blank*	*leave blank*	*leave blank*
Protocol	*leave blank*	*leave blank*	*leave blank*	*leave blank*
Scan Timing				
# of Echoes	1	1	1	1
TE	17	minful	17	minful
TR	500	34	500	34
Echo Train Length/Flip Angle	4	1	4	1
Bandwidth	15.63	15.63	15.63	15.63
Additional Parameters				
no entries required in this area				
Acquisition Timing				
Freq	256	256	256	256
Phase	256	256	256	256
NEX	1	1	1	1
Phase FOV	1.00	1.00	1.00	1.00
Preq DIR	S/I	S/I	S/I	S/I
Auto Center Freq	Peak	Peak	Peak	Peak
Autoshim	On	On	On	On
Phase Correct	On	On	On	On
Contrast	Off	Off	Off	Off
# of Reps B4 Pause	0	0	0	0
Scanning Range				
FOV	38	38	38	38
Slice Thickness	3	3	3	3
Spacing	1.5	1.5	1.5	1.5

	I/S Center	P/A Center	R/L Center
Start	0	0	
End	0		
# Slices	1	**Table Delta**	0.0

3-2-5 SNR Image Analysis for Eclipse

Image Analysis

Regions of interest in both signal and noise images can be measured directly in the image browser. Click the user interface button **Measure**, select the square or rectangular shape, and adjust its size and orientation when the shape is displayed in the selected image. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image.

Examples of typical intermediate Receiver Images for 4CH_NVARRAY are shown in *Figure 5* Typical composite and noise images for 4CH_NVARRAY are shown in *Figures 6* and *7*.

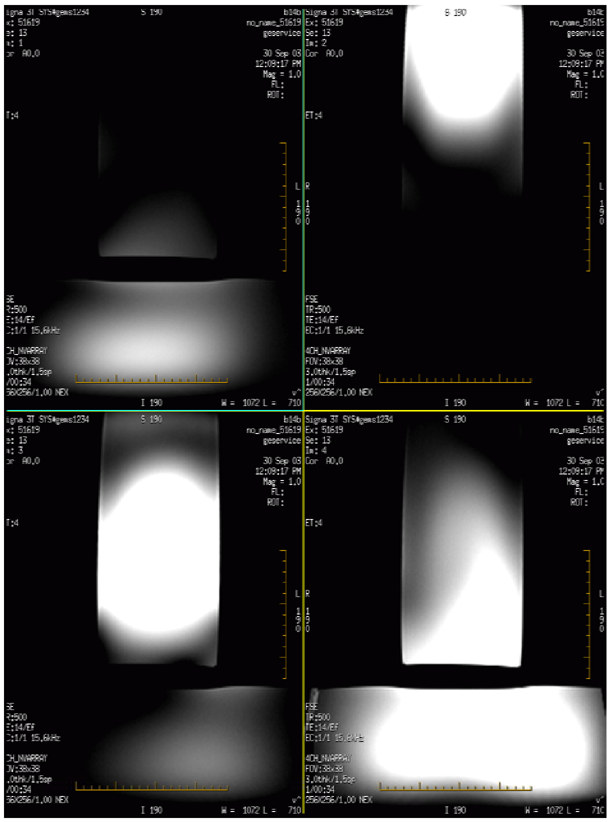


Figure 5: 4CH_NVARRAY intermediate receiver images 1-4.

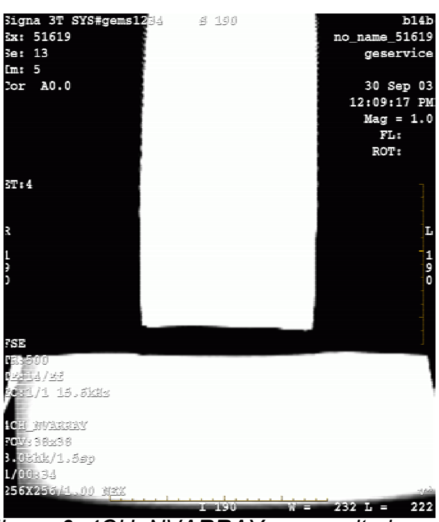


Figure 6: 4CH_NVARRAY composite image.

Figure 7: 4CH_NVARRAY noise image.

SNR Measurement

To acquire the composite signal image, take two SNR measurements for US4NVFULL. A 20cm height x 12cm width rectangular ROI should be used for the head section elements of the coil (see *Figure 8*). For the chest section of the coil a 30cm height x 10cm width rectangular ROI should be used (see *Figure 8*).

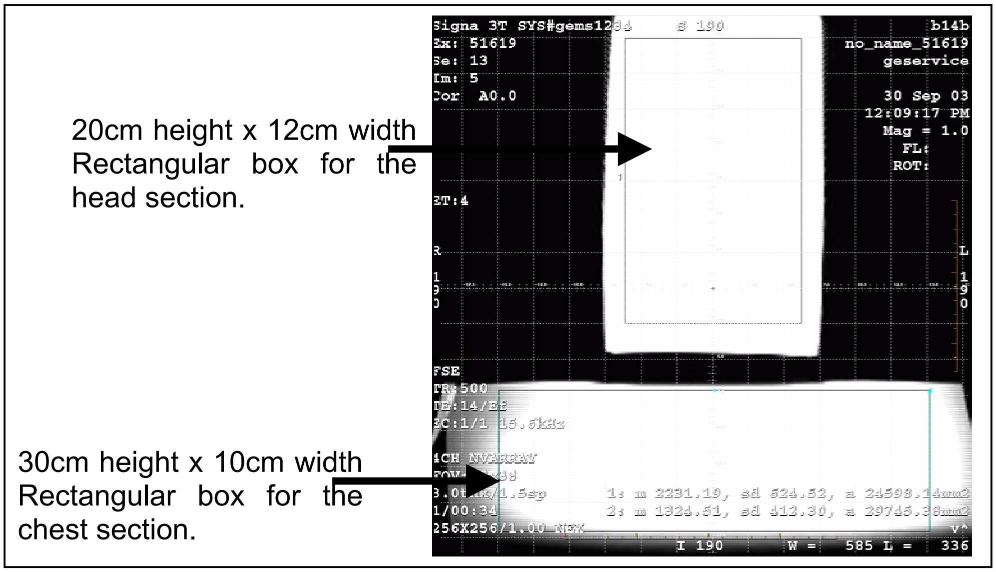


Figure 8: Composite 4CH_NVARRAY ROIs.

For the noise image, a 35cm^2 ROI should be used (see *Figure 9*).

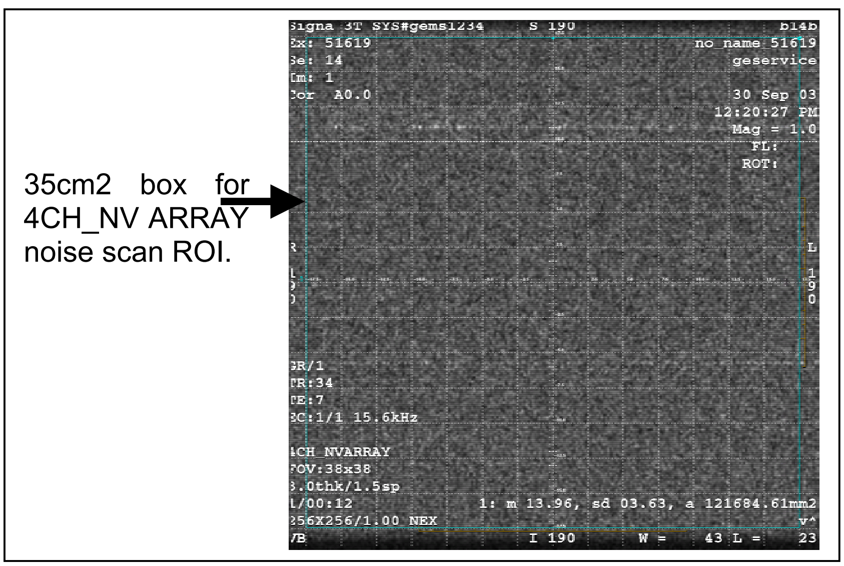


Figure 9: 4CH_NVARRAY noise ROI.

The SNR is calculated using the signal to noise ratios of the individual receiver channels:

$$\mathbf{SNR}_{\mathbf{Composite}} = \frac{\textbf{Mean of Signal within ROI}}{\textbf{Standard Deviation of Noise within ROI}}$$

Note: The SNR calculation uses the **MEAN** of the signal image and **STANDARD DEVIATION** of the noise image. SNR is measured on the composite image.

SNR Specification

The SNR measurements must be greater than or equal to the following specifications:

SNR SPECIFICATIONS – TABLE 3-2-5

Coil Element	Composite SNR
Head Section	410
Chest Section	220

3-2-6 SNR Image Analysis for EXCITE

Image Analysis

Regions of interest in both signal and noise images can be measured directly in the image browser. Click the user interface button **Measure**, select the square or rectangular shape, and adjust its size and orientation when the shape is displayed in the selected image. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image.

Examples of typical intermediate Receiver Images for US4NVFULL are shown in *Figure 10* Typical composite and noise images for US4NVFULL are shown in *Figures 11* and *12*.

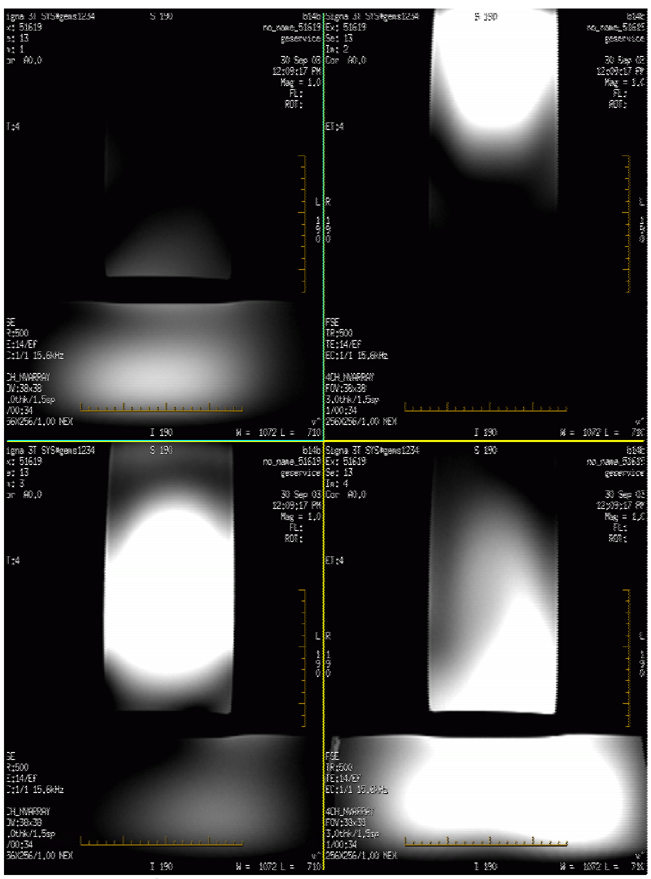


Figure 10: US4NVFULL intermediate receiver images 1-4.

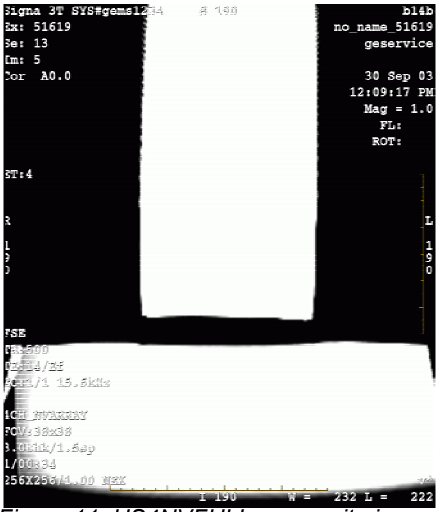


Figure 11: US4NVFULL composite image.

Figure 12: US4NVFULL noise image.

SNR Measurement

To acquire the composite signal image, take two SNR measurements for US4NVFULL. A 20cm height x 12cm width rectangular ROI should be used for the head section elements of the coil (see *Figure 13*). For the chest section of the coil a 30cm height x 10cm width rectangular ROI should be used (see *Figure 13*).

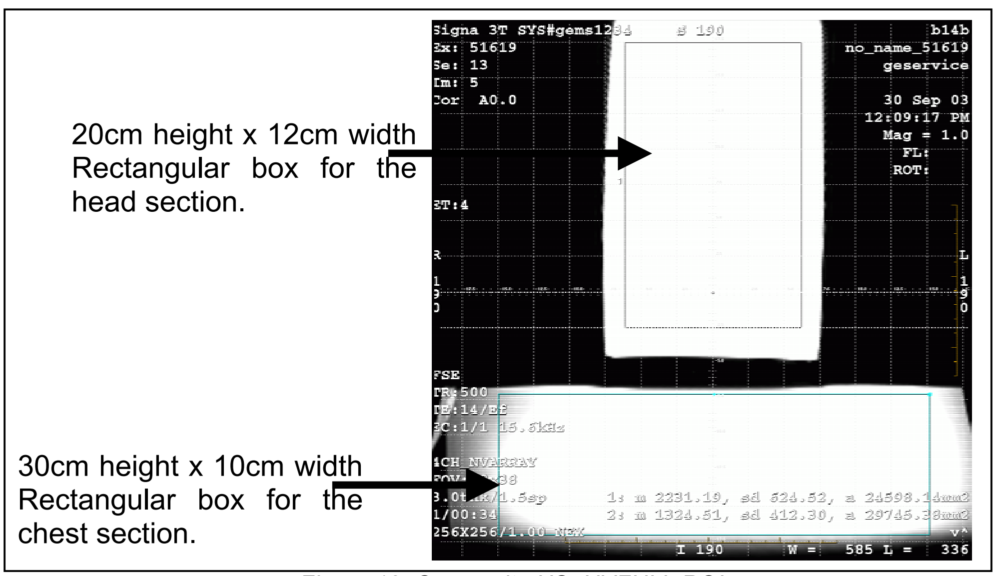


Figure 13: Composite US4NVFULL ROIs.

For the noise image, a $35cm^2$ ROI should be used (see *Figure 14*).

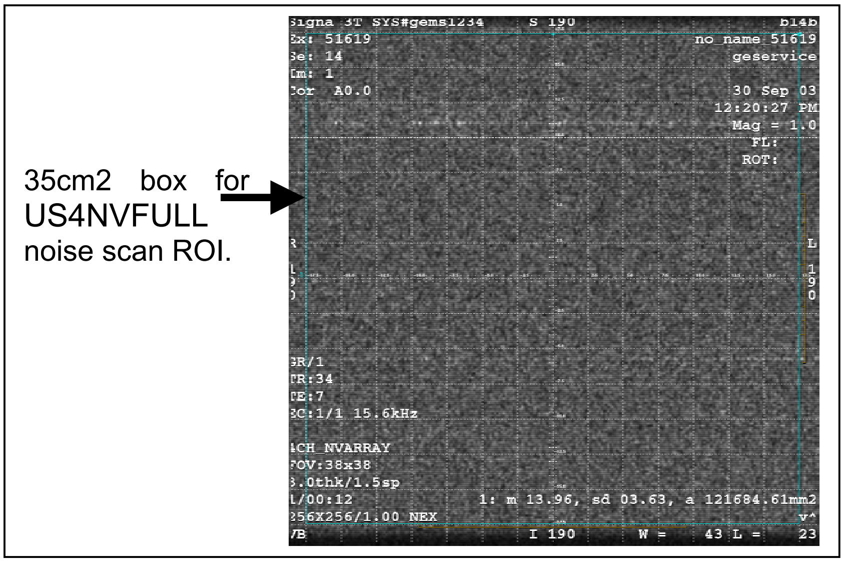


Figure 14: US4NVFULL noise ROI.

The SNR is calculated using the signal to noise ratios of the individual receiver channels:

$$\mathbf{SNR}_{\mathbf{Composite}} = \frac{\textbf{Mean of Signal within ROI}}{\textbf{Standard Deviation of Noise within ROI}}$$

Note: The SNR calculation uses the **MEAN** of the signal image and **STANDARD DEVIATION** of the noise image. SNR is measured on the composite image.

SNR Specification

The SNR measurements must be greater than or equal to the following specifications:

SNR SPECIFICATIONS – TABLE 3-2-6

Coil Element	Composite SNR
Head Section	410
Chest Section	220

3-3 External Cable Check

Check for continuity in the output cable.

Step 1 – Remove the cable cover from the bottom of the posterior section of the coil.

Step 2 – Unplug the cable from the coil.

Step 3 – Measure the continuity for the corresponding center pins and grounds.

Step 4 – Refer to *Figure 15* and Center Pin and Ground Connections – Table 3-3 for the correct pins for the Eclipse cable.

CENTER PIN AND GROUND CONNECTIONS – TABLE 3-3

System	Pin:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Eclipse																
Row A		G	NC	G	NC	G	NC	G	NC	G	G	G	NC	G	NC	G
Row B		S1	NC	S2	NC	S3	NC	S4	NC	S5	G	S6	NC	S7	NC	S8

(G = Ground, S1 = RF Signal Input for Channel 1, NC =No Connection, etc.)

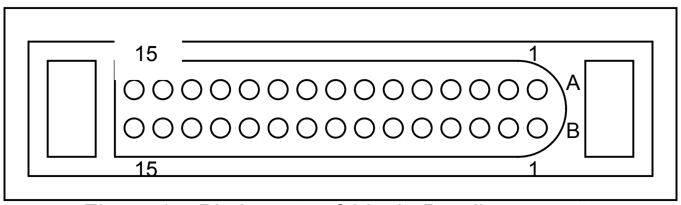


Figure 15: Pin layouts of 30-pin Bendix connector.

Step 5 – Refer to *Figure 16* for the location of the RF connectors for the EXCITE system.

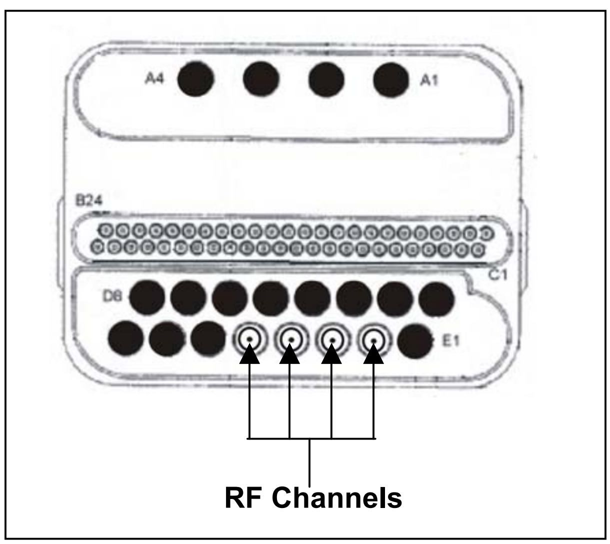


Figure 16: RF connectors for the EXCITE system.

Step 6 – Replace cable and cable cover.

3-4 PIN Diodes Check

Step 1 – Remove the coil covers.

Step 2 – Gently pry up the four white screw covers located down the middle of the anterior cover and the three white screw covers located down the middle of the posterior cover.

Step 3 – Remove all the screws found under the covers.

Step 4 – Remove the center covers from both the anterior and posterior coils.

Step 5 – Check the pin diodes located on the feed boards of the individual coil elements with a DVM.

Step 6 – If a diode is found to be either open or shorted:

- Take note of the diode's cathode-anode orientation.
- Carefully de-solder the surface mounted diodes from the feed board.
- Cut the leads of a new diode to 1/8".
- Place the new diode into the hole on the PCB. Align the leads of the diode with the surface mount pads.

The cathode pad is designated with a dot near the pad (see *Figure 17* for diode mounting).

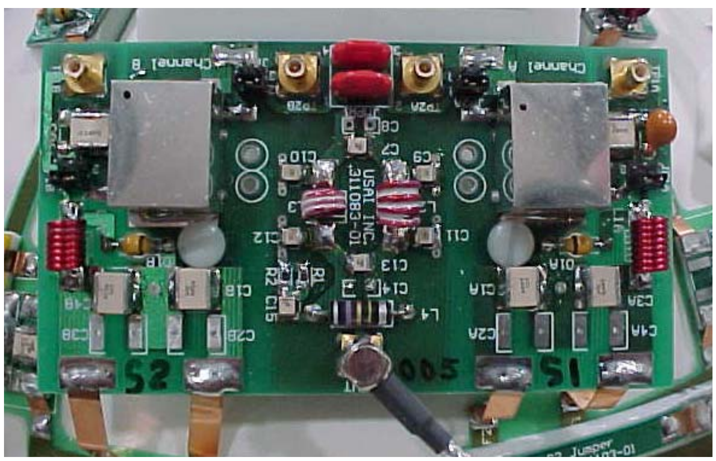


Figure 17: Diode mounting.

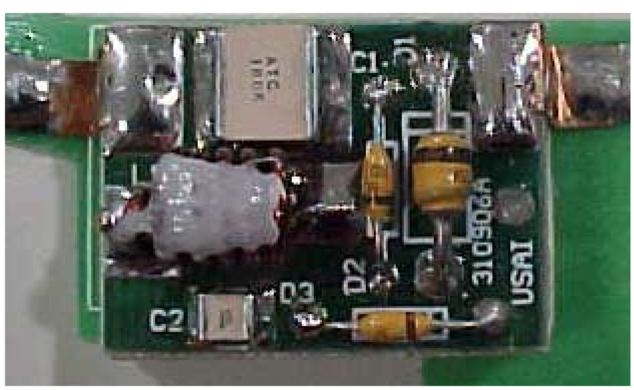


Figure 18: Passive decoupling PIN diode mounting.

3-5 Mechanical Hardware Check

None.

3-6 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again. Make sure the indicator light on the patient table turns green with the proper coil selected on the console.
One or more PIN diodes have failed.	From inside the coil, check the PIN diodes using a DVM. Use diode mode for forward-bias measurements and resistance mode for reverse-bias measurements. Note that there should be no reverse-bias leakage. Bad PIN diodes are usually shorted or exhibit reverse-bias leakage current.	Replace the coil.
There is a DC short or open somewhere within the DC path inside the coil.	Check the coil and associated DC paths with a DVM, in both diode check and resistance modes.	If a short or open is observed, check the output cable by itself (see below). If the cable and PIN diodes are okay, then replace the coil.
The output cable has a short or an open.	Disconnect the cable at the coil and check each coaxial line with a DVM (resistance). Try moving and twisting the cable as you watch the meter to find any intermittent connections. Inspect the center pins of the SMB and Bendix connections for wear and proper engagement into the mating connection.	Replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar 2-channel coil to see if the same problem occurs	Correct MRI system problem.

Coil ID is applicable only on the EXCITE system.

Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same. Try swapping preamplifiers between channels to see if the problem follows the preamp. If it does, then the preamplifier is bad.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise standard deviation measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the holder. Try running the scan without the phantom holder to see if the ghosting diminishes. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

Coil ID is applicable only on the EXCITE system.

3-7 Reversing Coil Polarity

This section applies only to the EXCITE system.

Step 1 – Remove the cable cover from the bottom of the posterior section of the coil.

Step 2 – Unplug the cable from the coil.

Step 3 – Remove the cover from the posterior section of the former.

Step 4 – Locate the three feed boards in the posterior section (see *Figure 19*).

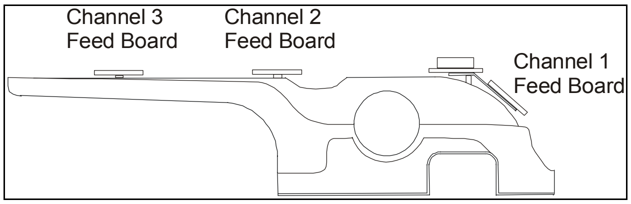


Figure 19: Feed board location.

Step 5 – Locate Red jumpers near the end of the board on the Channel 1 Feed Board.

Step 6 – Change the orientation of the Red shorting jumpers. Refer to *Figure 20* for before orientations, and *Figure 21* for after orientations.

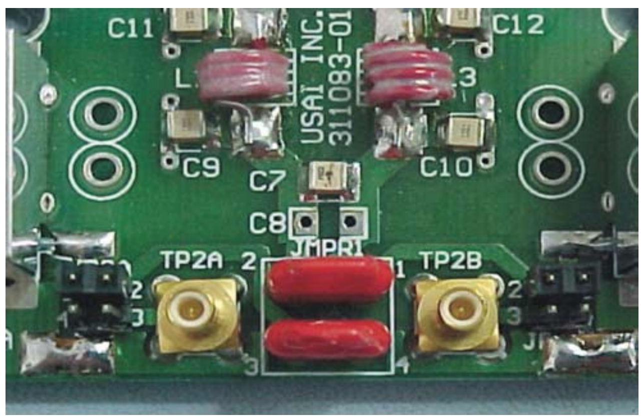


Figure 20*: Anti-quadrature orientation.*

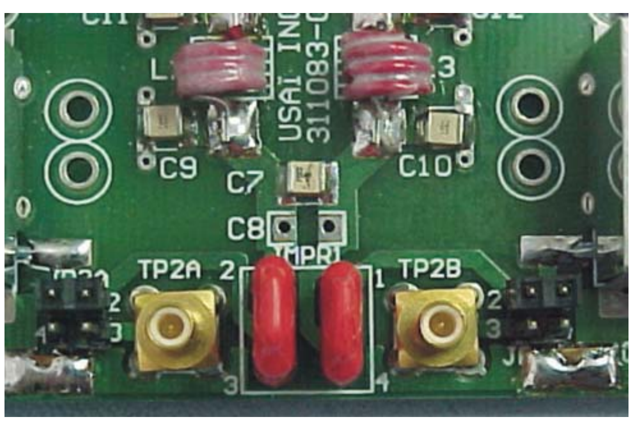


Figure 21: *Quadrature orientation.*

Step 7 - Locate Red jumpers near the end of the board on the Channel 2 Feed Board.

Step 8 – Change the orientation of the Red shorting jumpers. Refer to *Figure 21* for before orientations and *Figure 20* for after orientations.

Step 9 - Locate Red jumpers near the end of the board on the Channel 3 Feed Board.

Step 10 – Change the orientation of the Red shorting jumpers. Refer to *Figure 21* for before orientations and *Figure 20* for after orientations.

Step 11 – Reinstall the cover of the posterior section.

Step 12 – Reinstall the output cable.

Step 13 – Replace the cable cover.

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach the coil connector from the system before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.

CAUTION

Do not spray or pour cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronic components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

4-3 Carrying the Coil

The coil should be supported from underneath using both hands. Do not carry the coil by the cable.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

Step 1 – Remove cable cover on posterior side of coil.

Step 2 – Hold cable strain relief securely while unplugging connectors J1 thru J4.

Step 3 – Remove the cable.

Step 4 – Slide strain relief of the new cable into slot (see *Figure 22*).

Step 5 – Plug connectors J1 through J4 into corresponding jacks A1 through A4 (see *Figure 23*).

Step 6 – Replace the cable cover.

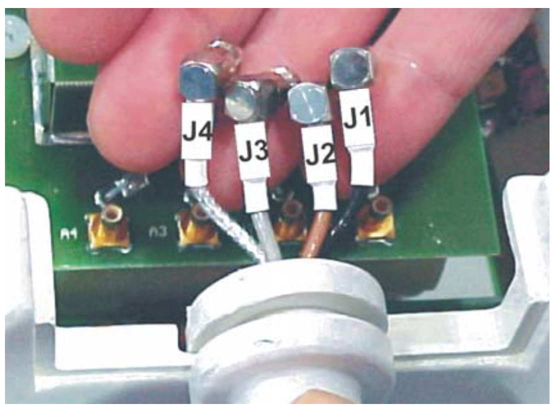


Figure 22: Attaching the new cable.

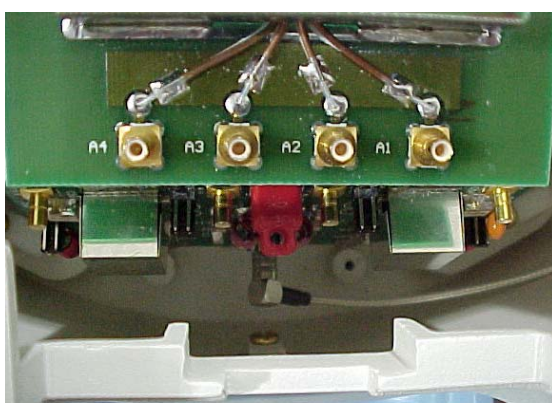


Figure 23: Jacks A1 through A4.

5-2 Mechanical Hardware Replacement

None.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units for Eclipse

ECLIPSE FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #
Coil	2383477-2
Cable Assembly	2383477-9
Phantom Kit	2383477-3
Phantom Positioner	2383477-4

6-2 Field Replaceable Units for EXCITE

EXCITE FIELD REPLACEABLE UNITS – TABLE 6-2

Part Name	GE Part #
Coil	2383478-2
Cable Assembly	2383478-5
Phantom Kit	2383477-3
Phantom Positioner	2383477-4

6-3 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES – TABLE 6-3

Part Name	GE Part #
Patient Comfort Pad	2383477-5
Wedge Pads	2383477-6

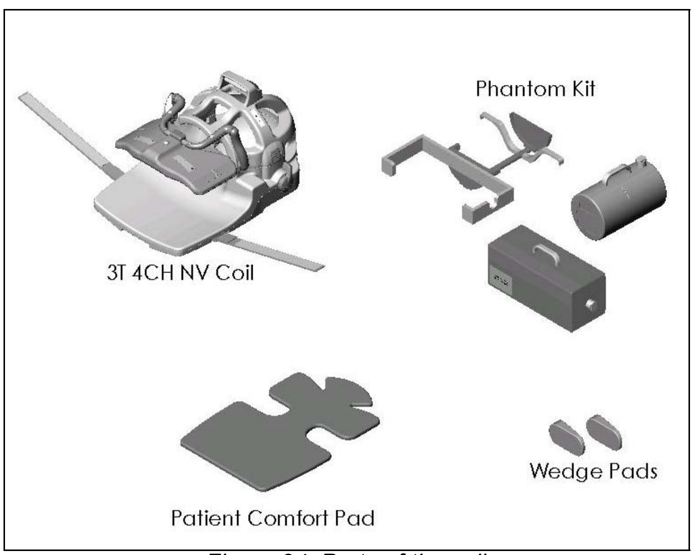


Figure 24: Parts of the coil.

SECTION 7 – APPENDIX

7-1 SNR Data Sheet

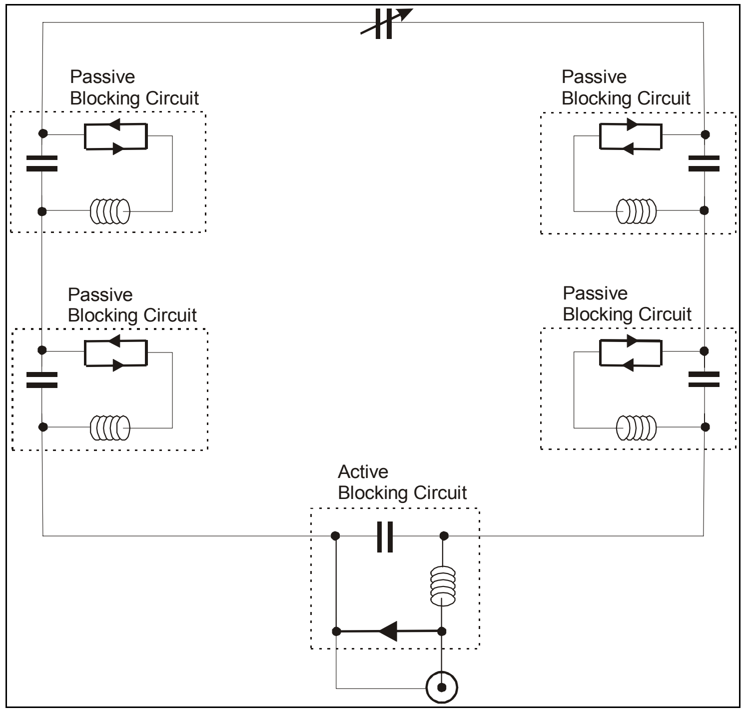
Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section. The composite SNR specifications are using the 4CH_NVARRAY mode on the Eclipse system.

Date	Comments						
Mode	R1	R2	TG	Signal Mean	Noise Std Dev	SNR	Spec Limit
Head Section							
Chest Section							

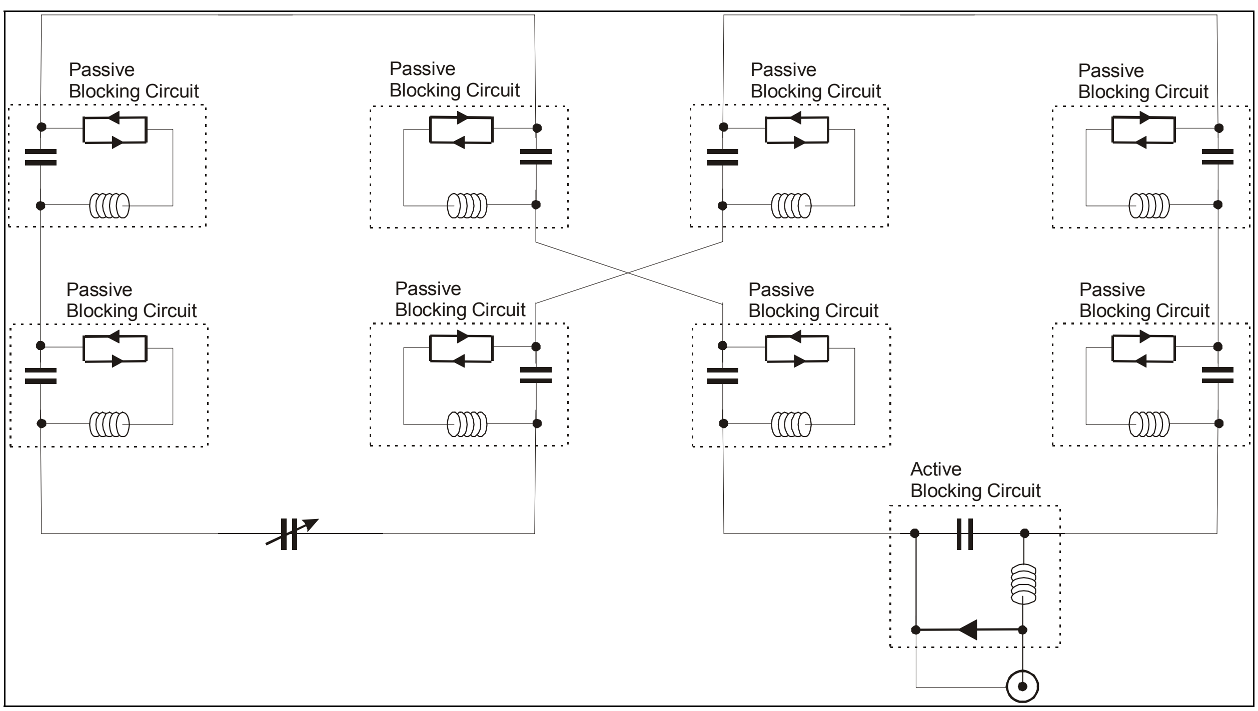
Date	Comments						
Mode	R1	R2	TG	Signal Mean	Noise Std Dev	SNR	Spec Limit
Head Section							
Chest Section							

Date	Comments						
Mode	R1	R2	TG	Signal Mean	Noise Std Dev	SNR	Spec Limit
Head Section							
Chest Section							

7-2 Schematic



Schematic for loops 1 and 2.



Schematic for saddles 1 through 5.

7-3 Coil Configuration

Parameter	Modes	
Coil Name	4CH_NVHEAD	4CH_NVARRAY
Coil Type	3	3
Extremity Coil	no	no
Cable Loss	1.3	1.3
Coil Loss	0.975	1.2
Recon Scale Factor	1.00	1.00
Linear vs. Quadrature	1	1
Multiple Receiver Coil?	yes	yes
Number of Receivers	2	4
Starting Receiver ID	1	0
Ending Receiver ID	2	3
Multi-Coil Port Enable	3	7
Multi-Coil Port Error Enable	3	7
Additional Transmit Attenuation	0	0
Number of Fast Receivers	0	0
Starting Fast Receiver ID	4	4
Ending Fast Receiver ID	4	4
Start TA Value	90	90
Start RG Value	12	12
Phased Array T/R Coil for Autoshim	-1	-1
Multi-Coil Recon Enable	0	0
Head Default Freq Direction	1	1
SCIC Axial	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100
SCIC Sagittal	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100
SCIC Coronal	1.5 0.35 0.2 5.0 128 0.2 1.05 100	1.5 0.35 0.2 5.0 128 0.2 1.05 100
Body Percent Exposed	25	25
aps1_mod	0	0
aps1_plane	1	1
aps1_mod_fov	200	200
opthickPS_mod	5	5
aps1_mod_PStloc	0	0
aps1_mod_PSrloc	0	0
Coil Transmit Position	4	2

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	12/03	Teresa DeMarco	First Issue
2	4/05	Pei H. Chan	Updated the SNR spec. in Tables 3-2-5 and 3-2-6